

Chest Pain

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Chest pain is a very common reason people seek medical care. Pain may be sharp or dull, although some people with a chest disorder describe their sensation as discomfort, tightness, pressure, gas, burning, or aching. Sometimes people also have pain in the back, neck, jaw, upper part of the abdomen, or arm. Other symptoms, such as nausea, cough, or difficulty breathing, may be present depending on the cause of the chest pain.

Many people are well aware that chest pain is a warning of potential life-threatening disorders and seek evaluation for minimal symptoms. Other people, including many with serious disease, minimize or ignore its warnings.

Causes of Chest Pain

Many disorders cause chest pain or discomfort. Not all of these disorders involve the heart. Chest pain may also be caused by disorders of the digestive system, lungs, muscles, nerves, or bones.

Common causes

Overall, the most common causes of chest pain are

- Disorders of the ribs, rib cartilage, chest muscles (musculoskeletal chest wall pain), or nerves in the chest
- Inflammation of the membrane that covers the lungs ([pleuritis](#))
- Inflammation of the membrane that covers the heart ([pericarditis](#))
- Digestive disorders (such as [esophageal reflux](#) or [esophageal spasm](#), [ulcer disease](#), or [gallstones](#))

- Heart attack or angina ([acute coronary syndromes](#) and [stable angina](#))
- Undiagnosed causes that go away on their own

[Acute coronary syndromes](#) (heart attack or unstable angina) involve a sudden blockage of an artery in the heart (coronary artery) that cuts off the blood supply to an area of the heart muscle. If some of the heart muscle dies because it does not get enough blood, that effect is termed a heart attack (myocardial infarction). In [stable angina](#), long-term narrowing of a coronary artery (for example by [atherosclerosis](#)) limits blood flow through that artery. This limited blood flow causes chest pain when people exert themselves.

Life-threatening causes

Some causes of chest pain are immediately life threatening but, except for heart attack or unstable angina, are less common:

- Heart attack or unstable angina
- A tear in the wall of the aorta (thoracic [aortic dissection](#))
- A type of collapsed lung in which pressure builds up enough to obstruct blood flow returning to the heart ([tension pneumothorax](#))
- A [rupture of the esophagus](#)
- Blockage of an artery to the lungs by a blood clot ([pulmonary embolism](#))

Other causes range from serious, potential threats to disorders that are simply uncomfortable.

Evaluation of Chest Pain

People with chest pain should be evaluated by a doctor. The following information can help people decide when evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with chest pain or discomfort, certain symptoms and characteristics are cause for concern. They include

- Crushing or squeezing pain
- Shortness of breath
- Sweating
- Nausea or vomiting

- Pain in the back, neck, jaw, upper abdomen, or one of the shoulders or arms
- Light-headedness or fainting
- Sensation of rapid or irregular heartbeat

When to see a doctor

Although not all causes of chest pain are serious, because some causes are life threatening, the following people should seek care in an emergency department right away:

- Those with new chest pain
- Those who have a warning sign
- Those who suspect that a heart attack is occurring (for example, because symptoms resemble a previous heart attack)

These people should call emergency services (911) or be taken to an emergency department as quickly as possible. People should not try to drive themselves to the hospital.

Chest pain that lasts for seconds (less than 30 seconds) is rarely caused by a heart disorder. People with very brief chest pain need to see a doctor, but emergency services are usually not needed.

People who have had chest pain for a longer time (a week or more) should see a doctor as soon as possible, but they should go to the hospital right away if they develop warning signs or the pain has steadily been getting worse or coming more often.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history and then do a physical examination. What they find during the history and physical examination often suggests a cause of the chest pain and the tests that may need to be done.

However, symptoms due to dangerous and not dangerous chest disorders overlap and vary greatly. For example, although a typical heart attack can cause dull, aching, squeezing, or crushing chest pain, some people with a heart attack have only mild chest discomfort or indigestion or arm or shoulder pain (referred pain—see figure

[What Is Referred Pain](#)

). On the other hand, people with indigestion may simply have an upset stomach, and those with shoulder pain may have only sore muscles. Similarly, although the chest is tender when touched in

people with musculoskeletal chest wall pain, the chest can also be tender in people who are having a heart attack. Thus, doctors usually do tests on people with chest pain.



Some Causes and Features of Chest Pain

Causes	Common Features*	Tests†
Heart disorders		
<u>Heart attack</u> (myocardial infarction) or <u>unstable angina</u>	Immediately life threatening	
	Dull, aching, squeezing, or crushing pain, sometimes sudden that	ECG, done several times over a period of time
	• Spreads to the jaw or arm • May be constant or come and go	Blood tests to measure substances that indicate heart damage (cardiac biomarkers)
	Sometimes shortness of breath or nausea	If ECG and cardiac biomarker levels are normal, often CT of heart arteries or a stress test
	Pain that occurs during exertion and is relieved by rest (angina pectoris)	If ECG or cardiac biomarker levels are abnormal, heart catheterization
	Certain abnormal heart sounds, heard through a stethoscope	
	Often warning signs‡	
<u>Thoracic aortic dissection</u> (a tear in the wall of the part of aorta in the chest)	Immediately life threatening	Chest x-ray studies
	Sudden, tearing pain that spreads to or starts in the middle of the back	An imaging study of the aorta
	Sometimes light-headedness, stroke, or pain, coldness, or numbness in a leg	• CT scan of the aorta • Transesophageal echocardiography (ultrasound study of the heart with the ultrasound)



MSD

Thoracic aortic

dissection (a tear in the wall of the part of aorta in the chest)

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